



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 25-26

Section: [EE]

Project Report On

Project Name: *Your NxtHome*

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Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)	Total =	
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or	Demonstrates limited understanding of verification processes and	Shows a basic understanding of verification processes and attempts to	Identifies and verifies system data, ensuring proper functional	Exhibits an exceptional understanding of verification processes and

	functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based	Limited understanding of graphical	Shows a basic understanding of creating user	Effectively identifies and meet the	Exhibits an exceptional work design

	application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .
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YOUR NXTHOME

Project Documentation

Case Study | ER Diagram | Database Schema

Making Renting Easy for Everyone

yournxthome@gmail.com | 09608-059184 | Dhaka, Bangladesh

SECTION 1: CASE STUDY

1.1 Project Overview

Project Name: Your Nxthome

Project Type: Web-based Rental Accommodation Platform

Your Nxthome is a user-friendly web and desktop application designed to simplify the rental accommodation search process in Bangladesh. The platform directly connects tenants — including bachelors, job holders, and families — with property owners, eliminating the need for door-to-door searching and reducing the friction that exists in traditional house-hunting.

1.2 Problem Statement

In Bangladesh, particularly in Dhaka, finding rental accommodation is a significant challenge. The following pain points were identified:

- Landlords tend to prefer family tenants over bachelor or single occupants, making it difficult for job holders and students.
- No centralized platform exists where tenants can filter listings based on tenant type (bachelor/family) or restriction type.
- Tenants must rely on word-of-mouth, paper notices, or exhaustive door-to-door visits.
- Landlords have no easy way to list, manage, and communicate with potential tenants.
- No transparent rating or feedback mechanism for landlords exists to help future tenants.

1.3 Proposed Solution

Your NxtHome addresses these challenges through a centralized digital platform with the following features:

- Structured property listings with filters for tenant type (Bachelor/Family) and restriction type (Restricted/Restriction-less).
- User registration and role-based access control (User, Admin, Developer).
- In-app messaging and offer-making between tenants and property owners.
- Booking management with Pending, Confirmed, and Rejected states.
- Wishlist feature to save preferred properties.
- Star-based rating and comment system for property owners.
- Admin panel for platform-level oversight and management.
- Developer panel for database access, SQL runner, activity logs, and full statistics.

1.4 Target Users

User Type	Description	Key Actions
Regular User (Tenant)	Bachelors, students, job holders, families looking for rent	Browse posts, booking, Wishlist, chat, make offers, rate owners
Property Owner	Users who list their properties for rent	Create posts, manage bookings, accept/reject requests, chat
Admin	Platform administrators	Manage users, posts, bookings, messages; view platform overview
Developer	Technical team	DB browser, SQL runner, activity logs, full statistics, user role management
Guest	Unauthenticated visitors	Browse posts (view-only), cannot book/chat/rate/Wishlist

1.5 System Modules

Module 1: Authentication & User Management

- Sign Up: Users register with Name, Contact, Address, NID, Email, Occupation, UserID, Password, and Profile Photo.
- Sign In: Authenticated login with **show/hide password feature**. Redirects to **role-specific dashboard**.
- Forgot Password: **Multi-step verification** using Username, NID, and Contact number before resetting.
- **Role-based Dashboard**: Separate interfaces for User, Admin, and Developer roles.

Module 2: Post Management

- Property owners can create listings with: Post Title, Location, Rent, Description, Tenant Type, Restriction Type, and Photo.
- All published posts are visible on the View Posts page.
- Admin can delete or edit any post from the Admin Panel.
- Posts can be **searched and filtered** by Tenant Type (Bachelor/Family) and Restriction Type.

Module 3: Booking System

- Tenants can book a property from the Post Details view.
- Bookings go into a Pending state and appear in the owner's Booking Requests panel.
- Owners can Accept or Reject requests; tenants are notified via auto-message.
- Confirmed bookings appear in both tenant and owner dashboards.
- Admin can view, approve, reject, or delete all bookings platform-wide.

Module 4: Messaging & Offers

- Tenants can initiate a **direct chat with a property owner** from the post details page.
- Make an Offer feature sends a formatted offer message (e.g., [OFFER] ₹17000 for Post #1).
- Messages are visible to Admin and Developer for moderation purposes.

Module 5: Wishlist

- Users can add any post to their personal Wishlist for later reference.
- Wishlist shows: Post ID, Title, Location, Rent, Owner, and Date Added.
- Items can be removed individually or viewed in detail.

Module 6: Ratings

- After a confirmed booking, **tenants can rate the property owner** (1–5 stars) with an optional comment.
- Ratings are visible on post cards and in user profiles.
- Users can view all their given ratings under My Ratings section.

Module 7: Admin Panel

- Dashboard Overview: Total Users, Posts, Bookings (Pending/Confirmed/Rejected), Messages, Ratings.
- All Posts: Search, view, edit, or delete any listing.
- User Info: Add, update, delete users (**Admin role cannot change user roles — only Developer can**).
- All Bookings: View all bookings with status; Approve, Reject, or Delete.
- All Messages: View all platform messages with timestamps; Delete messages.

Module 8: Developer Panel

- DB Browser: View any database table in real-time; delete rows, clear tables, **export CSV**.
- SQL Runner: Execute SELECT, INSERT, UPDATE, DELETE queries directly.
- Full Stats: Detailed breakdown of users, posts, bookings, messages, and ratings.
- Activity Logs: Recent bookings, messages, and post **activity log**.
- User Info: Full user management including **role assignment (User/Admin)**.

1.6 Key Features Summary

Feature	Available To	Description
Guest Browsing	Anyone	View all listings without logging in
Post Filtering	Guest / User	Filter by Bachelor/Family and Restricted/Free
Booking	Signed-in Users	Request to book a flat; owner approves/rejects
Wishlist	Signed-in Users	Save favorite listings
Chat & offers	Signed-in Users	Direct messaging and rent negotiation
Ratings	Users with confirmed bookings	1–5-star rating with comment for owner
Create Listing	Signed-in Users	Post flat/room with full details and photo
Admin Management	Admin Role	Manage users, posts, bookings, messages
Developer Access	Developer Role	Full DB access, SQL runner, activity logs

1.7 Technology Stack

Layer	Technology
Frontend	Windows Forms / Desktop UI (C# / WinForms)
Backend	C# .NET
Database	Microsoft SQL Server (T-SQL)
Authentication	Custom username/password with NID verification
File Storage	Local file system (ImagePath stored in DB)

1.8 System Actors and Use Case Summary

Actor	Key Use Cases
Guest	Browse posts, filter listings, view post details
Registered User (Tenant)	Sign up/in, view/filter posts, book, Wishlist, chat, offer, rate owner, manage profile
Property Owner (User)	Create/publish posts, manage booking requests (accept/reject), chat with tenants
Admin	Manage all users, posts, bookings, messages; view platform statistics
Developer	Full database control, SQL execution, activity monitoring, user role management, Manage all users

1.9 Constraints and Assumptions

- All monetary values are in BDT (Bangladeshi Taka ট).
- **NID (National ID) is used as a unique identifier for password reset verification.**
- **Only users with confirmed bookings can rate a property owner.**
- A regular Admin cannot change a user's role — only **a Developer can assign Admin/User roles.**
- Photos are stored as file paths or binary data in the database.
- The system currently supports desktop/web interface — no native mobile app.

1.10 Benefits and Impact

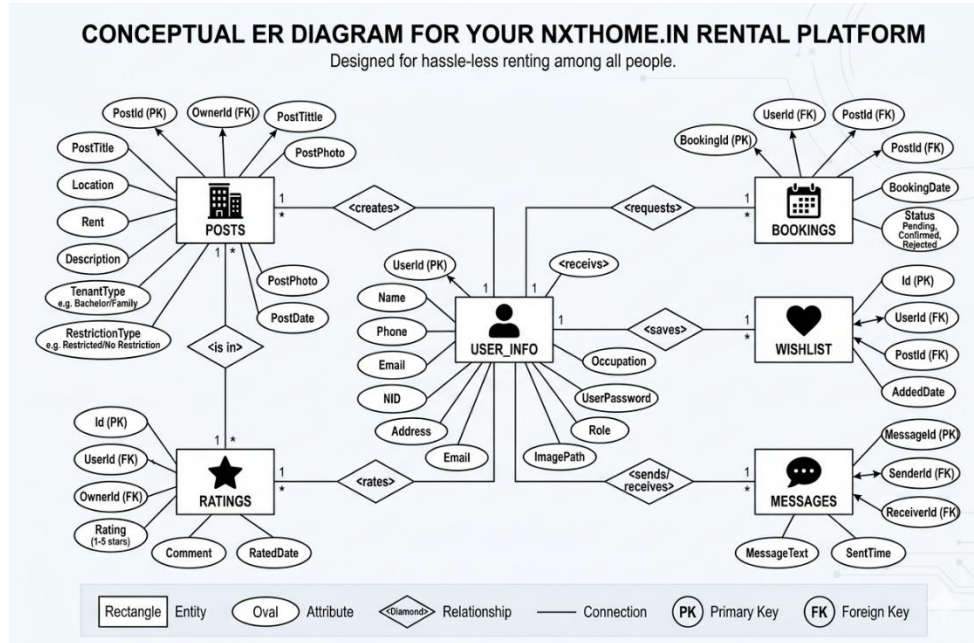
- Reduces the average house-hunting time from weeks to hours.
- Provides bachelors and job holders with equal visibility in the rental market.
- Creates a transparent, review-driven ecosystem for property owners.
- Centralizes the entire rental process — search, communicate, negotiate, book — on a single platform.
- Empowers administrators with real-time control and analytics over the platform.

SECTION 2: ENTITY-RELATIONSHIP (ER) DIAGRAM

2.1 Entities and Attributes

The following entities form the core of the Your NxtHome data model:

ER Diagram



Entity: user_info

Attribute	Data Type	Constraint	Description
UserId	NVARCHAR(50)	PRIMARY KEY, NOT NULL	Unique user identifier / username
Name	NVARCHAR(100)	NOT NULL	Full name of the user
Phone	NVARCHAR(20)	NOT NULL	Contact phone number
Email	NVARCHAR(100)	NOT NULL	Email address
NID	NVARCHAR(50)	NOT NULL	National ID for identity verification
Address	NVARCHAR(255)	NULL	Residential address
Occupation	NVARCHAR(100)	NULL	Job / occupation type
UserPassword	NVARCHAR(255)	NOT NULL	Hashed password
Role	VARCHAR(20)	NULL	User role: User / Admin / Developer
ImagePath	NVARCHAR(225)	NULL	Path to profile photo

Entity: Posts

Attribute	Data Type	Constraint	Description
PostId	INT IDENTITY	PRIMARY KEY	Auto-incremented post identifier
OwnerId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	Owner of the listing
PostTitle	NVARCHAR(200)	NOT NULL	Title of the property listing
Location	NVARCHAR(200)	NOT NULL	Physical location of the property
Rent	INT	NOT NULL	Monthly rent amount in BDT
Description	NVARCHAR(1000)	NULL	Detailed description of the property
TenantType	VARCHAR(20)	NULL	Bachelor or Family
RestrictionType	VARCHAR(20)	NULL	Restricted or RestrictionLess
PostPhoto	VARBINARY(MAX)	NULL	Binary image data of the property
PostDate	DATETIME	DEFAULT GETDATE()	Date the post was published

Entity: Bookings

Attribute	Data Type	Constraint	Description
BookingId	INT IDENTITY	PRIMARY KEY	Auto-incremented booking ID
UserId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	The tenant making the booking
PostId	INT	FK → Posts.PostId, NOT NULL	The property being booked
BookingDate	DATETIME	DEFAULT GETDATE()	Date of booking request
Status	VARCHAR(20)	DEFAULT 'Pending'	Pending / Confirmed / Rejected

Entity: Wishlist

Attribute	Data Type	Constraint	Description
Id	INT IDENTITY	PRIMARY KEY	Auto-incremented wishlist entry ID
UserId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	User who saved the post
PostId	INT	FK → Posts.PostId, NOT NULL	The saved post
AddedDate	DATETIME	DEFAULT GETDATE()	Date added to wishlist

Entity: Ratings

Attribute	Data Type	Constraint	Description
Id	INT IDENTITY	PRIMARY KEY	Auto-incremented rating ID
UserId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	User (tenant) who gave the rating
OwnerId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	The property owner being rated
Rating	INT	CHECK (1-5)	Star rating from 1 to 5
Comment	NVARCHAR(500)	NULL	Optional review comment
RatedDate	DATETIME	DEFAULT GETDATE()	Date the rating was submitted

Entity: Messages

Attribute	Data Type	Constraint	Description
MessageId	INT IDENTITY	PRIMARY KEY	Auto-incremented message ID
SenderId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	User who sent the message
ReceiverId	NVARCHAR(50)	FK → user_info.UserId, NOT NULL	User who received the message
MessageText	NVARCHAR(1000)	NOT NULL	Content of the message
SentTime	DATETIME	DEFAULT GETDATE()	Timestamp of the message

2.2 Entity Relationships

Relationship	From Entity	To Entity	Cardinality	Description
OWNS / LISTS	user_info	Posts	One-to-Many (1:N)	One user can own many posts
BOOKS	user_info	Bookings	One-to-Many (1:N)	One user can make many bookings
IS BOOKED	Posts	Bookings	One-to-Many (1:N)	One post can have many booking requests
WISHLISTS	user_info	Wishlist	One-to-Many (1:N)	One user can have many wishlist entries
IS WISHLISTED	Posts	Wishlist	One-to-Many (1:N)	One post can be wishlisted many times
RATES (as rater)	user_info (Tenant)	Ratings	One-to-Many (1:N)	A tenant can give ratings to many owners
IS RATED (as owner)	user_info (Owner)	Ratings	One-to-Many (1:N)	An owner can receive many ratings
SENDS	user_info	Messages	One-to-Many (1:N)	One user can send many messages
RECEIVES	user_info	Messages	One-to-Many (1:N)	One user can receive many messages

SECTION 3: DATABASE SCHEMA

3.1 Schema Overview

The Your NxtHome database consists of 6 tables in Microsoft SQL Server. All tables follow relational normalization principles (3NF). Foreign key constraints ensure referential integrity across all entities.

Table Name	Primary Key	Foreign Keys	Purpose
user_info	UserId (NVARCHAR)	None	Stores all user accounts and roles
Posts	PostId (INT IDENTITY)	OwnerId → user_info	Property listings created by owners
Bookings	BookingId (INT IDENTITY)	UserId → user_info, PostId → Posts	Booking requests by tenants
Wishlist	Id (INT IDENTITY)	UserId → user_info, PostId → Posts	Saved/favorited posts by users
Ratings	Id (INT IDENTITY)	UserId → user_info, OwnerId → user_info	Star ratings from tenants to owners
Messages	MessageId (INT IDENTITY)	SenderId → user_info, ReceiverId → user_info	Chat messages between users

3.2 Full Database Schema(Query) (SQL)

Table 1: user_info

```
CREATE TABLE user_info (      UserId      NVARCHAR(50) NOT NULL PRIMARY KEY,      Name
NVARCHAR(100) NOT NULL,      Phone      NVARCHAR(20) NOT NULL,      Email
NVARCHAR(100) NOT NULL,      NID      NVARCHAR(50) NOT NULL,      Address
NVARCHAR(255) NULL,      Occupation NVARCHAR(100) NULL,      UserPassword NVARCHAR(255) NOT
NULL,      Role      VARCHAR(20) NULL DEFAULT 'User',      ImagePath NVARCHAR(225)
NULL );
```

Table 2: Posts

```
CREATE TABLE Posts (      PostId      INT IDENTITY(1,1) PRIMARY KEY,      OwnerId
NVARCHAR(50) NOT NULL,      PostTitle NVARCHAR(200) NOT NULL,      Location
NVARCHAR(200) NOT NULL,      Rent      INT NOT NULL,      Description
NVARCHAR(1000) NULL,      TenantType VARCHAR(20) NULL, -- 'Bachelor' or 'Family'
RestrictionType VARCHAR(20) NULL, -- 'Restricted' or 'RestrictionLess'      PostPhoto
VARBINARY(MAX) NULL,      PostDate      DATETIME DEFAULT GETDATE(),      FOREIGN KEY
(OwnerId) REFERENCES user_info(UserId) );
```

Table 3: Bookings

```
CREATE TABLE Bookings (      BookingId      INT IDENTITY(1,1) PRIMARY KEY,      UserId
NVARCHAR(50) NOT NULL,      PostId      INT NOT NULL,      BookingDate DATETIME
DEFAULT GETDATE(),      Status      VARCHAR(20) DEFAULT 'Pending',      -- Status values:
```

```
'Pending', 'Confirmed', 'Rejected' FOREIGN KEY (UserId) REFERENCES user_info(UserId),
FOREIGN KEY (PostId) REFERENCES Posts(PostId) );
```

Table 4: Wishlist

```
CREATE TABLE Wishlist (      Id              INT IDENTITY(1,1) PRIMARY KEY,      UserId
NVARCHAR(50) NOT NULL,      PostId      INT      NOT NULL,      AddedDate  DATETIME
DEFAULT GETDATE(),      FOREIGN KEY (UserId) REFERENCES user_info(UserId),      FOREIGN KEY
(PostId) REFERENCES Posts(PostId) );
```

Table 5: Ratings

```
CREATE TABLE Ratings (      Id              INT IDENTITY(1,1) PRIMARY KEY,      UserId
NVARCHAR(50) NOT NULL, -- Tenant who rated      OwnerId      NVARCHAR(50) NOT NULL, --
Owner being rated      Rating      INT      CHECK (Rating BETWEEN 1 AND 5),      Comment
NVARCHAR(500) NULL,      RatedDate  DATETIME  DEFAULT GETDATE(),      FOREIGN KEY
(UserId) REFERENCES user_info(UserId),      FOREIGN KEY (OwnerId) REFERENCES
user_info(UserId) );
```

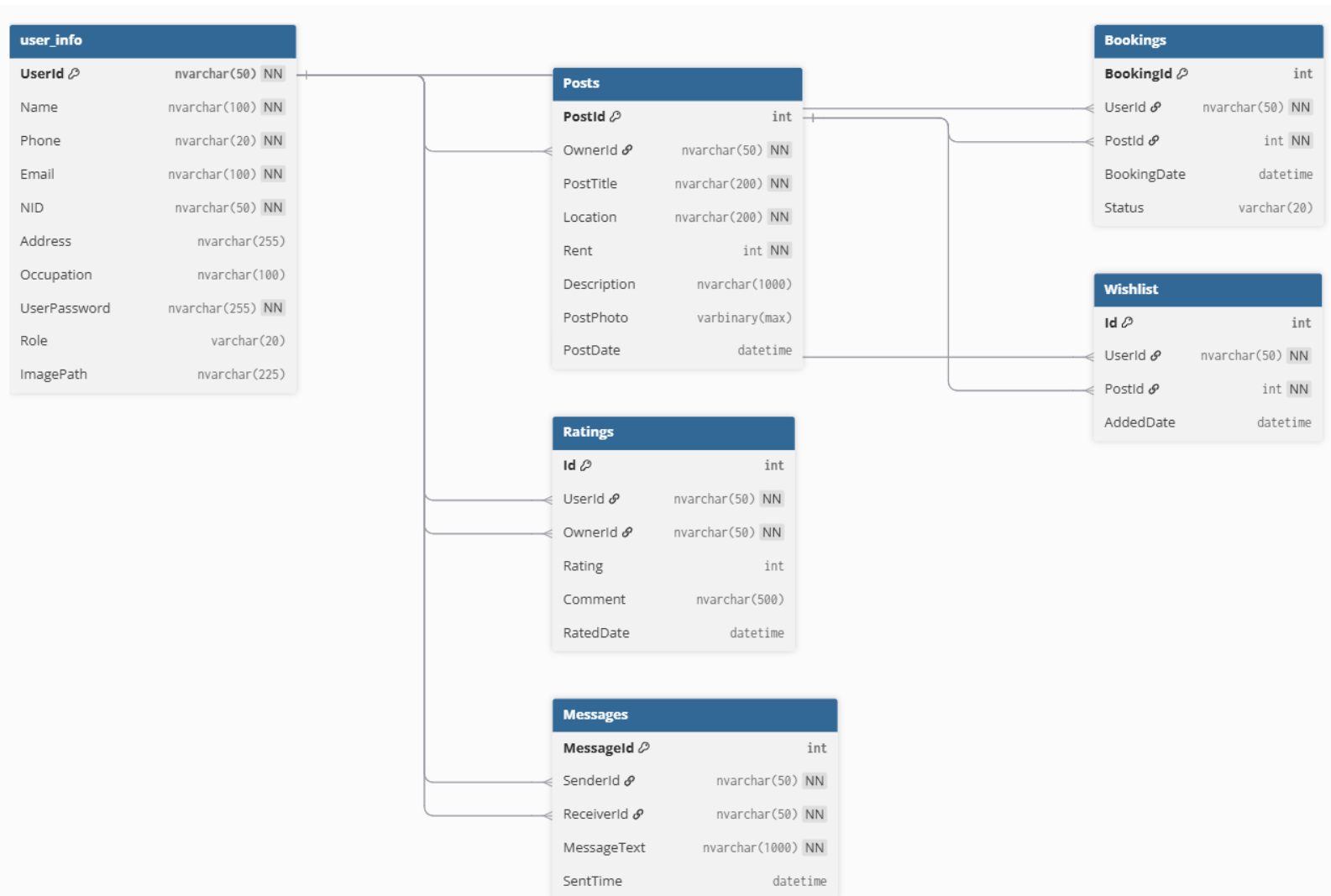
Table 6: Messages

```
CREATE TABLE Messages (      MessageId      INT IDENTITY(1,1) PRIMARY KEY,      SenderId
NVARCHAR(50) NOT NULL,      ReceiverId  NVARCHAR(50) NOT NULL,      MessageText
NVARCHAR(1000) NOT NULL,      SentTime  DATETIME  DEFAULT GETDATE(),      FOREIGN KEY
(SenderId) REFERENCES user_info(UserId),      FOREIGN KEY (ReceiverId) REFERENCES
user_info(UserId) );
```

3.3 Referential Integrity Summary

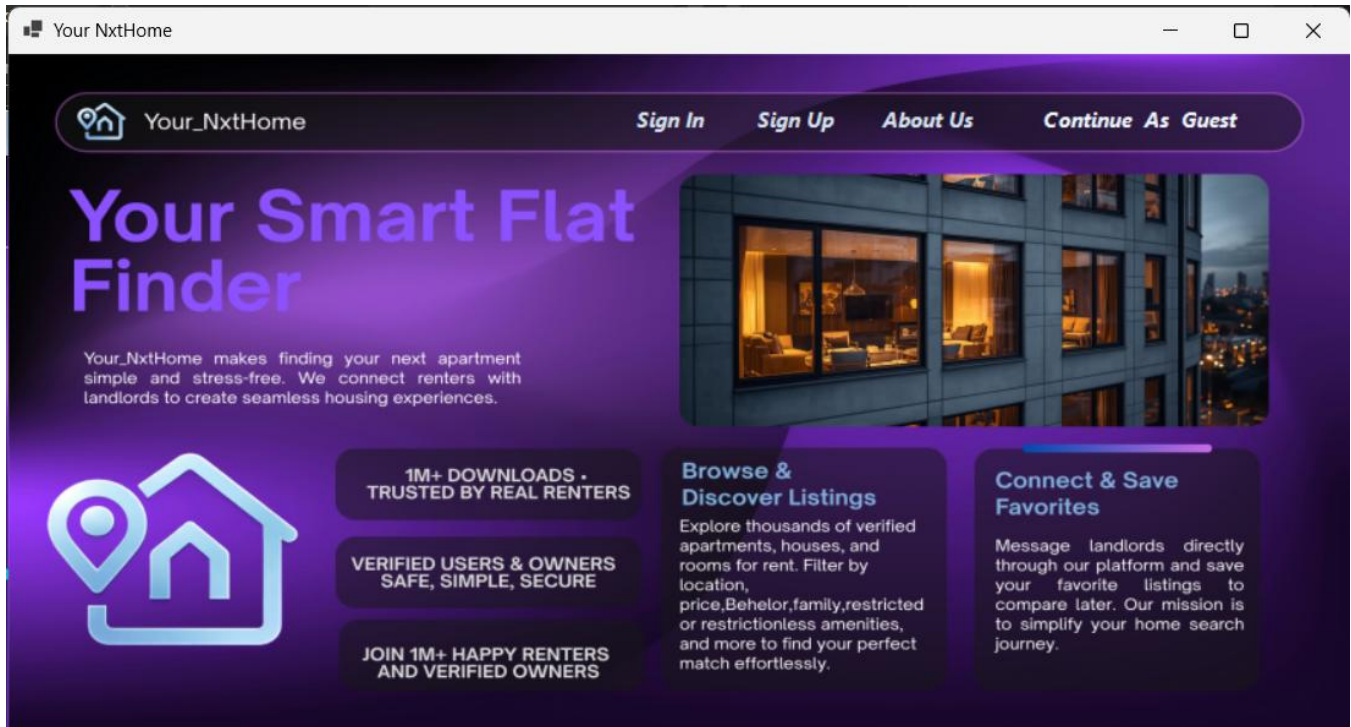
Child Table	Child Column	Parent Table	Parent Column	On Delete
Posts	OwnerId	user_info	UserId	Restrict (owner must exist)
Bookings	UserId	user_info	UserId	Restrict
Bookings	PostId	Posts	PostId	Restrict
Wishlist	UserId	user_info	UserId	Restrict
Wishlist	PostId	Posts	PostId	Restrict
Ratings	UserId	user_info	UserId	Restrict
Ratings	OwnerId	user_info	UserId	Restrict
Messages	SenderId	user_info	UserId	Restrict
Messages	ReceiverId	user_info	UserId	Restrict

DATABASE SCHEMA :



4. TRANSITION

HOMEPAGE



SIGN UP

The screenshot shows the sign-up form of the 'Your_NxtHome' application. The form is set against a dark purple background. It contains several input fields for user registration: 'Name', 'Contact', 'Address', 'NID', 'E-Mail', 'Occupation', 'Userid', and 'Password'. To the right of these fields is a large square placeholder for a profile picture, with an 'Upload Photo' button below it. At the bottom right of the form is a 'Save' button. A red 'Back' link is located in the bottom right corner of the form area.

SIGN IN

Open sign up
for registration

Your NxtHome

YOUR NEXTHOME

Username : niloy12

Password :

[Don't Have An Account?](#)

[Forgot Password](#)

[Show Password](#)

[Sign In](#)

[Back](#)

Open sign up
for registration

Sign Up

Full Name :
Email :
Phone :
Occupation :
Address :
Password :
Confirm Password :

[Sign Up](#)

SHOW PASSWORD

Password : 1234

[Hide Password](#)

PASSWORD RESET
WITH VERIFIED NID
& CONTACT

Your NxtHome

YOUR NEXTHOME

Password Reset

Username :

Contact :

Nid :

[Verify Info](#)

New Password :

[Update Password](#)

[Home](#)

[Back](#)

USER DASHBOARD

Your NxtHome - Dashboard

YOUR NEXTHOME



ID : niloy12

- [My Profile](#)
- [View Posts](#)
- [Wishlist](#)
- [Pending Booking](#)
- [Confirmed Booking](#)
- [My Ratings](#)
- [Messages](#)
- [Booking Requests](#)
- [Sign Out](#)

My Profile

Name : niloy

Email : niloy12@gmail.com

Phone : 123456789

Occupation : student

Address : kuril

NID : 123456789

[Edit Profile](#)

[Change Photo](#)

EDIT INFO

Your NxtHome - Dashboard

YOUR NEXTHOME

ID : niloy12

- My Profile
- View Posts
- Wishlist
- Pending Booking
- Confirmed Booking
- My Ratings
- Messages
- Booking Requests

Sign Out

My Profile

Name : niloy

Email : niloy12@gmail.com

Phone : 123456789

Occupation : student

Address : kuril

NID : 123456789

New User ID : niloy12

New Password :

Save Changes Change Photo

POST FILTERING

VIEW POST

SEARCH POSTS

Your NxtHome - View Posts

Search by title or location... Search Create Post Dashboard

Tenant: ☐ Bachelor ☐ Family Type: ☐ Restricted ☒ No Restriction

Single room for bechelor

₹ 9,000 / month

Bashundhara C block

Bachelor | Free

5.0 Robin

Details Book

2 bedroom flat available for family

₹ 19,500 / month

kuril

Family | Free

1.0 Karim

Details Book

2 bedroom flat available for family

kuril

₹ 19,500 / month

Owner: Karim

Rating: 1.0 / 5

Family | No Restriction

2 bedroom, 1 dining, 2 washroom

Book Now Add to Wishlist

Chat with Owner Make an Offer

CREATE POST

Your NxtHome - Create Post

Create New Post

Post Title :

Location :

Rent (₹/mo) :

Description :

Tenant Type : ☒ Bachelor ☐ Family

Restriction : ☒ No Restriction ☐ Restricted

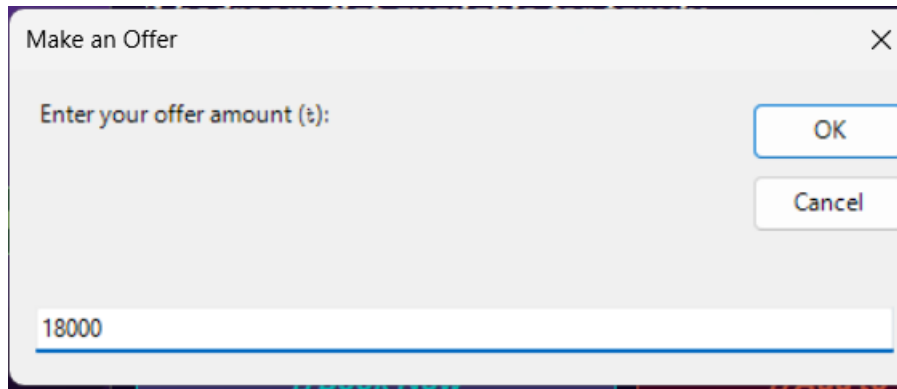
Photo Preview

No photo selected

Pick Photo

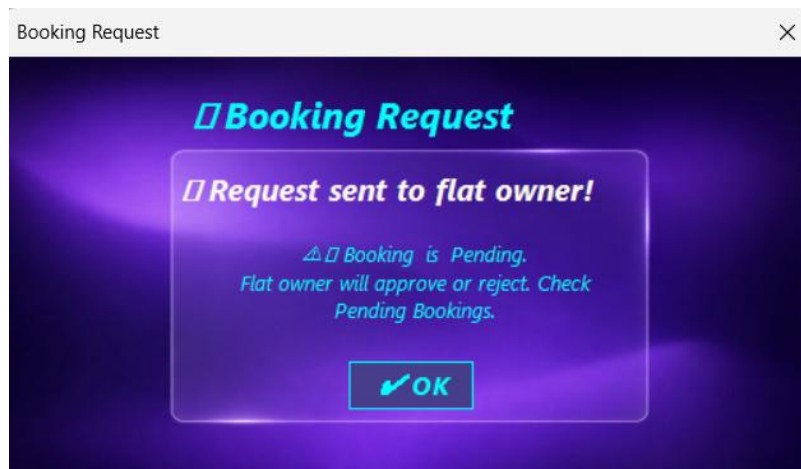
Publish Post Cancel

MAKE OFFER POP UP WINDOW



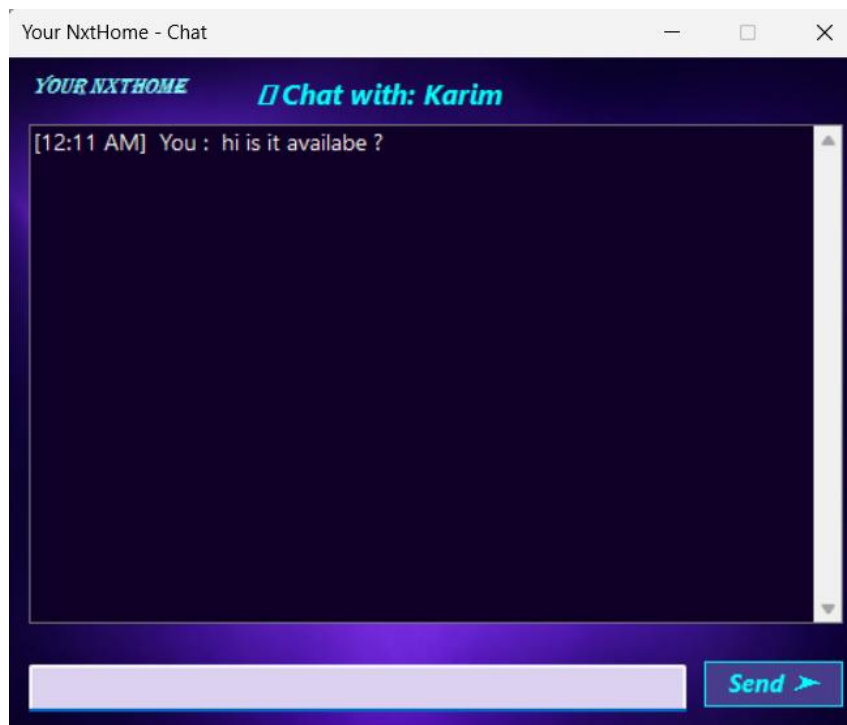
A screenshot of a web browser window showing a 'Make an Offer' pop-up. The pop-up has a title bar with 'Make an Offer' and a close button. The main content area contains the text 'Enter your offer amount (€):' followed by a text input field containing '18000'. To the right of the input field are two buttons: 'OK' and 'Cancel'.

BOOKING POP-UP



A screenshot of a web browser window showing a 'Booking Request' pop-up. The pop-up has a title bar with 'Booking Request' and a close button. The main content area has a dark blue background with a glowing effect. It features the text 'Booking Request' in a large, bold, white font, followed by 'Request sent to flat owner!' in a smaller, bold, white font. Below this, there is a message: 'Booking is Pending. Flat owner will approve or reject. Check Pending Bookings.' in a smaller, white font. At the bottom center, there is a button with a checkmark icon and the text 'OK'.

CHAT WITH OWNER



A screenshot of a web browser window showing a chat interface titled 'Your NxtHome - Chat'. The chat area has a dark blue background. At the top, it says 'YOUR NEXTHOME' and 'Chat with: Karim'. Below this, there is a message from 'You' at '[12:11 AM]' that says 'hi is it avaiable ?'. The chat area is mostly empty, with a scroll bar on the right. At the bottom, there is a text input field and a 'Send' button with a paper plane icon.

WISHLIST

Your NxtHome - Wishlist

♥️ My Wishlist

ID	Title	Location	Rent	Owner	Added
1	2 bedroom flat available f...	kuril	₹ 19500	Karim	5/7/2026

[Remove](#) [View Details](#) [← Dashboard](#)

PENDING BOOKING

Your NxtHome - Dashboard

YOUR NEXTHOME



ID : niloy12

- [My Profile](#)
- [View Posts](#)
- [♥️ My Wishlist](#)
- [⌂ Pending Booking](#)
- [✔ Confirmed Booking](#)
- [★ My Ratings](#)
- [Messages](#)
- [Booking Requests](#)

[Sign Out](#)

⌂ Pending Bookings

📄 2 bedroom flat available for family | kuril | 5/7/2026

CHECK MESSAGES & OFFER

The screenshot shows the 'My Messages' dashboard with a list of messages and an open chat window.

From / To	Last Message	Time
niloy	[OFFER] ₹18000 for Post #1	07 May 12:11 AM
Karim	✗ Sorry! Your booking request has been declined by...	24 Apr 10:00 PM
John	☑ Great news! Your booking request has been ACCEPT...	24 Apr 09:59 PM

Chat Window: Chat with: niloy

[12:11 AM] niloy : hi is it available ?
 [12:11 AM] niloy : [OFFER] ₹18000 for Post #1

Buttons: **Open Chat**, **Sign Out**, **Send**

BOOKING REQUEST

The screenshot shows the 'Pending Booking Requests' dashboard with a table of requests and a confirmation dialog.

ID	Renter Name	Post Title	Location	Booking Date
5	niloy	2 bedroom flat available for family	kuril	07 May 2026 12:12 AM

Confirm Dialog: Accept this booking request?
 Buttons: **Yes**, **No**

Buttons: **Accept**, **Reject**, **Refresh**, **Close**

CONFIRMED BOOKINGS

Your NxtHome - Dashboard

YOUR NEXTHOME



ID : niloy12

- My Profile
- View Posts
- Wishlist
- Pending Booking
- Confirmed Booking**
- My Ratings
- Messages
- Booking Requests

— Sign Out

✓ Confirmed Bookings

☒ 2 bedroom flat available for family | kuril | 5/7/2026

★ Rate Owner

RATE OWNER

Rate Owner

★ Rate Flat Owner

Select Owner :

Karim|K12

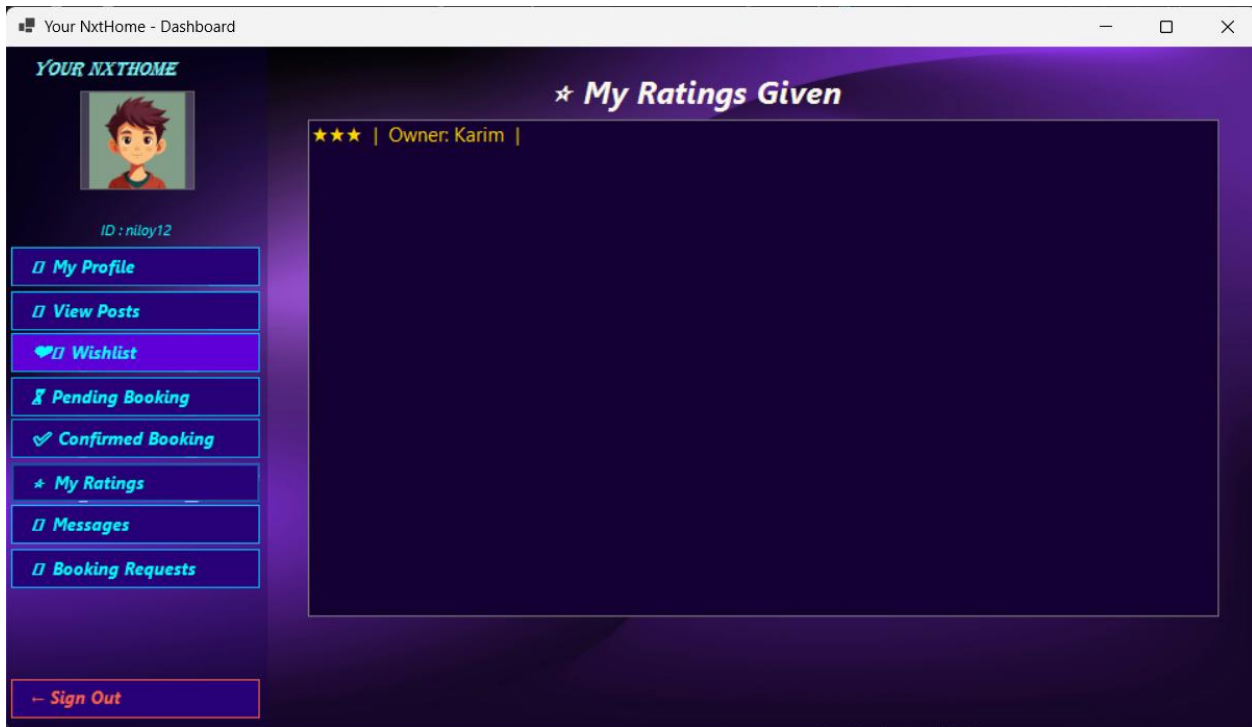
Your Rating : 3 / 5

★ ★ ★ ★ ★

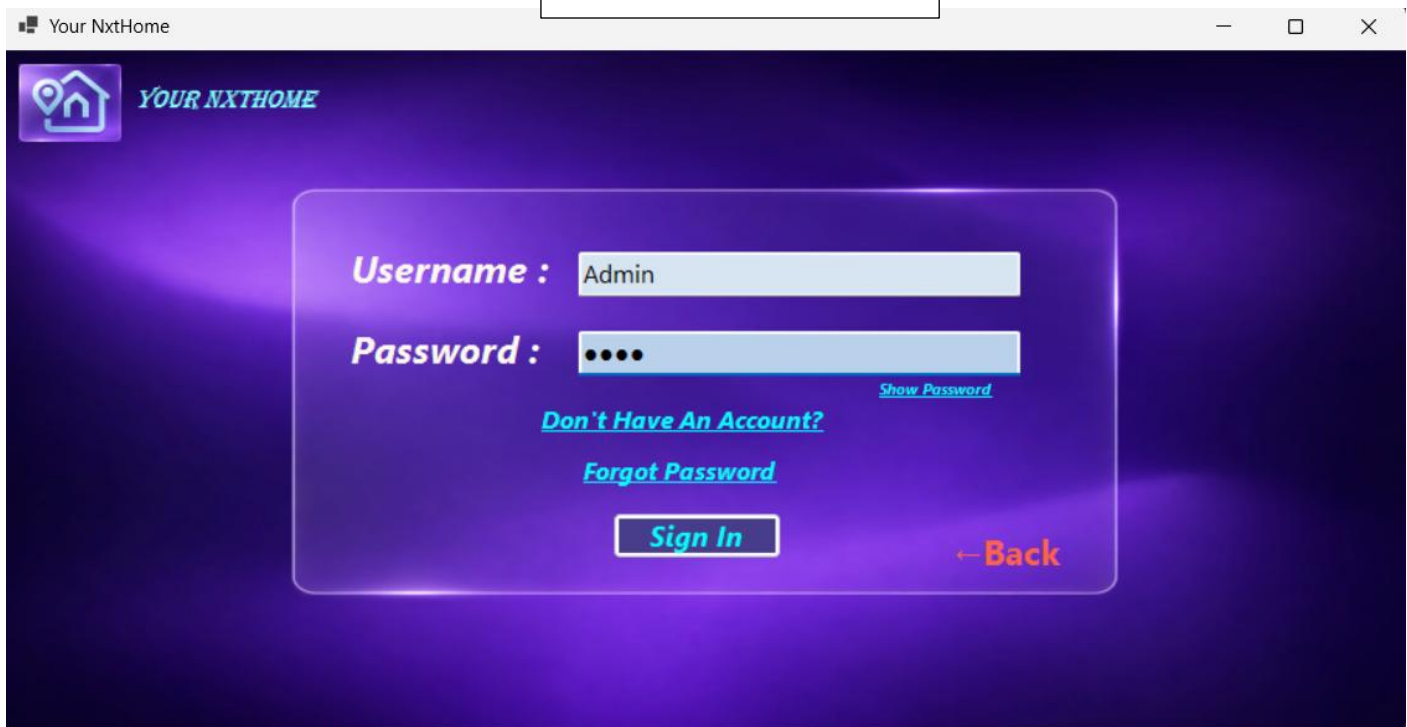
Comment (optional) :

★ Submit Rating Cancel

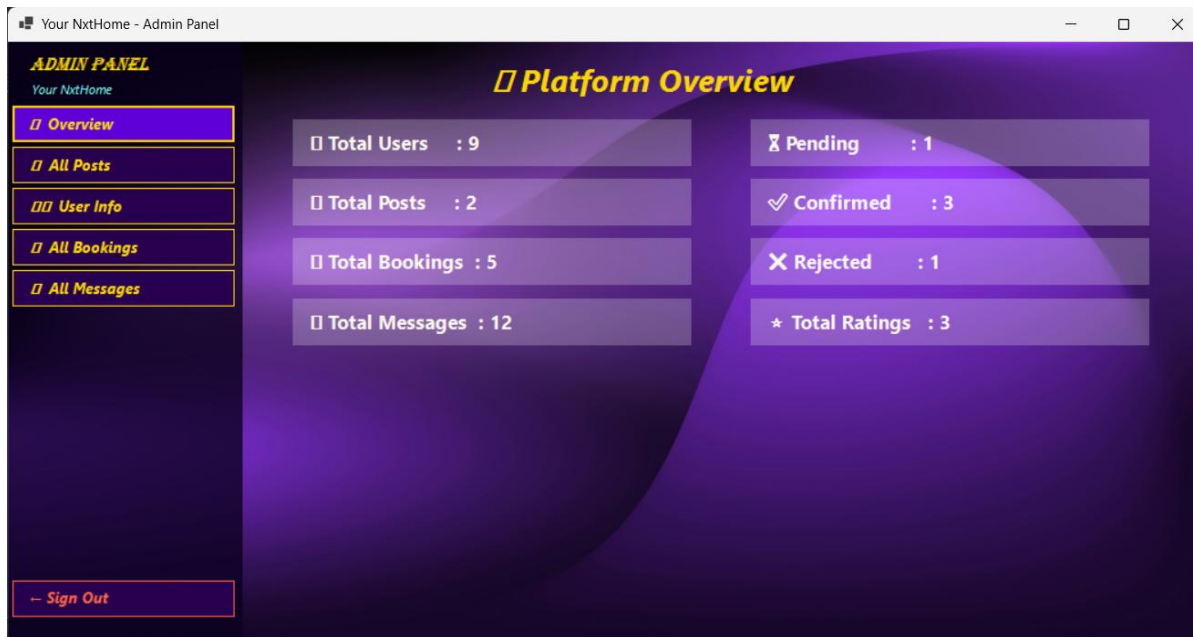
MY RATINGS



SIGN IN {BUT AS ADMIN}



OVERVIEW



ALL POSTS

The screenshot shows the 'All Posts' dashboard. The sidebar is identical to the Overview page. The main content area is titled 'All Posts' and features a search bar with the placeholder 'Search title / location / owner...' and a 'Search' button. Below the search bar is a table with the following data:

ID	Title	Owner	Location	Rent	Tenant	Type	Date
2	Single room for bechelor	Robin	Bashundhora C bl...	₹ 9000	Bachelor	RestrictionLess	4/23/
1	2 bedroom flat available ...	Karim	kuril	₹ 19500	Family	RestrictionLess	4/21/

At the bottom of the table, there are two buttons: 'Delete Post' and 'Edit Post'.

ADMIN CAN DELETE,
UPDATE POST
DETAILS

The screenshot shows the 'Edit Post (Admin)' form. It contains the following fields and options:

- Post Title :** 2 bedroom flat available for family
- Location :** kuril
- Rent (/mo) :** 19500
- Description :** 2 bedroom, 1 daing,2 washroom
- Tenant :** ☐ Bachelor ☒ Family
- Restriction :** ☒ No Restriction ☐ Restricted

At the bottom, there are two buttons: 'Save Changes' and 'Cancel'.

USER_INFO

User Info

Userid	Name	Phone	Email	NID	Address	Occupation	UserPassword
arafat	Arafat	012301230123	arafat@gmail.c...	112233445566	Baridhara	Job holder	arafat
demo	demo	1	demo@gmail.c...	1	demo	demo	demo
john12	John	12345678	John@gmail.co...	1234	bonsree	Student	1234
K12	Karim	789456123	karim@gmail.c...	85274196	mirpur	Student	1234

Name :

Occupation :

Add

Contact :

Userid :

Update

Address :

Password :

Delete

NID :

Role :

Clear

E-Mail :

Back

ALL BOOKINGS

Your NxtHome - Admin Panel

ADMIN PANEL

Your NxtHome

Overview

All Posts

User Info

All Bookings

All Messages

Sign Out

All Bookings

ID	Renter	Owner	Post	Location	Status	Date
5	niloy	Karim	2 bedroom flat available fo...	kuril	Confirmed	5/7/2026
4	montu	Robin	Single room for bechelor	Bashundhora C...	Pending	4/24/2026
3	John	Robin	Single room for bechelor	Bashundhora C...	Confirmed	4/23/2026
2	Karim	Karim	2 bedroom flat available fo...	kuril	Confirmed	4/23/2026
1	John	Karim	2 bedroom flat available fo...	kuril	Rejected	4/23/2026

Approve

Reject

Delete

ALL MESSAGES

Your NxtHome - Admin Panel

ADMIN PANEL

Your NxtHome

Overview

All Posts

User Info

All Bookings

All Messages

Sign Out

All Messages

ID	From	To	Message	Time
13	Karim	niloy	Great news! Your booking request has been ACCEPTED ...	07 May 12
12	niloy	Karim	[OFFER] ₳18000 for Post #1	07 May 12
11	niloy	Karim	hi is it avallabe ?	07 May 12
10	John	Robin	accepted	24 Apr 10:
9	Robin	John	only for you , i can 8500 last ,if you agree with me deal th...	24 Apr 10:
8	John	Robin	can you make it for me in 8000 per month?	24 Apr 10:
7	Karim	Karim	Sorry! Your booking request has been declined by the o...	24 Apr 10:
6	Karim	John	Great news! Your booking request has been ACCEPTED ...	24 Apr 09:
5	Robin	John	Great news! Your booking request has been ACCEPTED ...	23 Apr 10:
4	John	Robin	[OFFER] ₳8000 for Post #2	23 Apr 09:
3	Karim	John	ok done	23 Apr 12:
2	John	Karim	[OFFER] ₳17000 for Post #1	22 Apr 07:

Delete Message

SIGN IN {AS A DEVELOPER}

Your NxtHome

 YOUR NEXTHOME

Username :

Password : [Show Password](#)

[Don't Have An Account?](#)

[Forgot Password](#)

[← Back](#)

DB BROWSER

Developer Panel — Full Access

Database Browser

Select Table : Table: [Bookings] Rows: 5

BookinAId	UserId	PostId	BookinaDate	Status
1	john12	1	4/23/2026 12:15 AM	Rejected
2	k12	1	4/23/2026 12:16 AM	Confirmed
3	john12	2	4/23/2026 9:20 AM	Confirmed
4	montu	2	4/24/2026 9:55 PM	Pending
5	niloy12	1	5/7/2026 12:12 AM	Confirmed

[Refresh](#) [Delete Row](#) [Clear Table](#) [Export CSV](#)

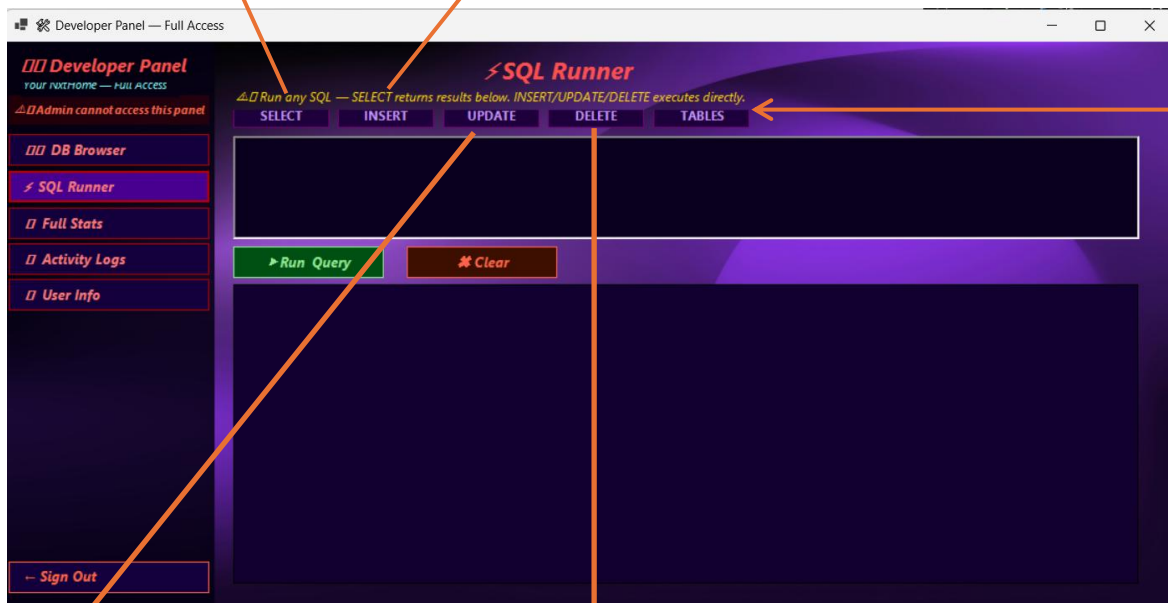
[Sign Out](#)

SQL RUNNER

```
INSERT INTO Posts (OwnerId, PostTitle, Location, Rent, Description, TenantType, RestrictionType, PostDate)  
VALUES ('userId', 'Title', 'Location', 10000, 'Description', 'Bachelor', 'RestrictionLess', GETDATE())
```

```
SELECT * FROM User_info
```

SET TEMPLATE
QUERY TO MAKE
MORE EASIER



```
DELETE FROM Posts WHERE PostId=1
```

```
UPDATE User_info SET Role='Admin' WHERE UserId='userId'
```

```
-- Check tables  
SELECT TABLE_NAME, TABLE_TYPE  
FROM INFORMATION_SCHEMA.TABLES  
ORDER BY TABLE_NAME
```

FULL STATS

Developer Panel
Your nickname — Full Access

Admin cannot access this panel

DB Browser
SQL Runner
Full Stats
Activity Logs
User Info

Sign Out Refresh Stats

Full Platform Statistics

DATABASE STATISTICS

- USERS**
 - Total Users : 9
 - Admin Users : 1
 - Regular Users : 8
- POSTS**
 - Total Posts : 2
 - Bachelor Posts : 1
 - Family Posts : 1
 - Restricted : 0
 - RestrictionLess : 2
- BOOKINGS**
 - Total Bookings : 5
 - Pending : 1
 - Confirmed : 3
 - Rejected : 1
- MESSAGES & RATINGS**
 - Total Messages : 12
 - Offer Messages : 2
 - Total Ratings : 3

ACTIVITY LOGS

Developer Panel
Your nickname — Full Access

Admin cannot access this panel

DB Browser
SQL Runner
Full Stats
Activity Logs
User Info

Sign Out Refresh Logs

Recent Activity Logs

RECENT ACTIVITY LOG

- LATEST BOOKINGS**
 - [07 May 12:12 AM] niloy → 2 bedroom flat available for family [Confirmed]
 - [24 Apr 09:55 PM] montu → Single room for bechelor [Pending]
 - [23 Apr 09:20 AM] John → Single room for bechelor [Confirmed]
 - [23 Apr 12:16 AM] Karim → 2 bedroom flat available for family [Confirmed]
 - [23 Apr 12:15 AM] John → 2 bedroom flat available for family [Rejected]
- LATEST MESSAGES**
 - [07 May 12:18 AM] Karim → niloy: Great news! Your booking request has b...
 - [07 May 12:11 AM] niloy → Karim: [OFFER] 8000 for Post #1
 - [07 May 12:11 AM] niloy → Karim: hi is it availabe ?
 - [24 Apr 10:14 PM] John → Robin: accepted
 - [24 Apr 10:13 PM] Robin → John: only for you , i can 8500 last ,if you a...
 - [24 Apr 10:12 PM] John → Robin: can you make it for me in 8000 per month...
 - [24 Apr 10:00 PM] Karim → Karim: Sorry! Your booking request has been d...
 - [24 Apr 09:59 PM] Karim → John: Great news! Your booking request has b...
 - [23 Apr 10:00 AM] Robin → John: Great news! Your booking request has b...
 - [23 Apr 09:48 AM] John → Robin: [OFFER] 8000 for Post #2
- LATEST POSTS**
 - [23 Apr 09:19 AM] Robin posted: Single room for bechelor (Bashundhora C block)
 - [21 Apr 07:16 PM] Karim posted: 2 bedroom flat available for family (kuril)

USER_INFO

Developer Panel
Your nickname — Full Access

Admin cannot access this panel

DB Browser
SQL Runner
Full Stats
Activity Logs
User Info

Sign Out Refresh Logs

User Info

Userid	Name	Phone	Email	NID	Address	Occupation	UserPassword
arafat	Arafat	012301230123	arafat@gmail.c...	112233445566	Baridhara	Job holder	arafat
demo	demo	1	demo@gmail.c...	1	demo	demo	demo
john12	John	12345678	John@gmail.co...	1234	bonsree	Student	1234
K12	Karim	789456123	karim@gmail.c...	85274196	mirpur	Student	1234

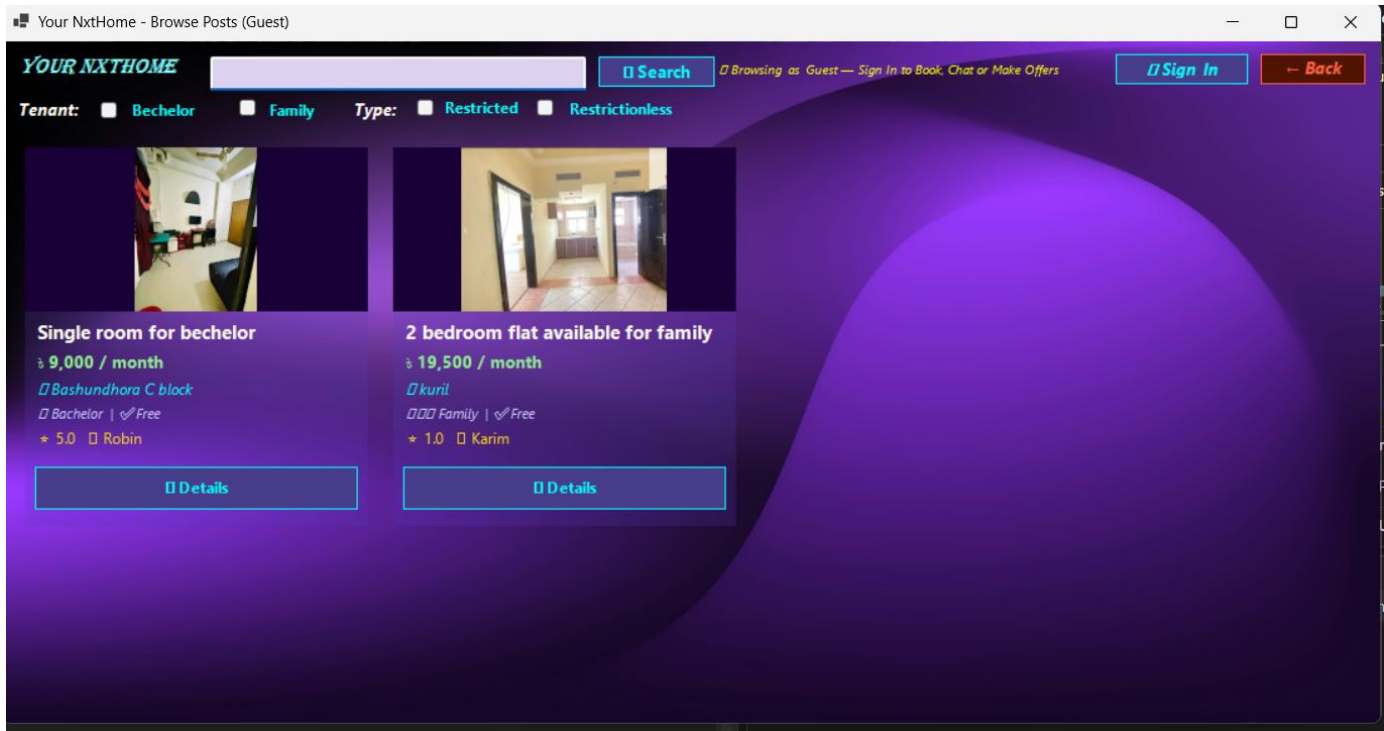
Name : Contact : Address : NID : E-Mail :

Occupation : Userid : Password : Role : User Admin

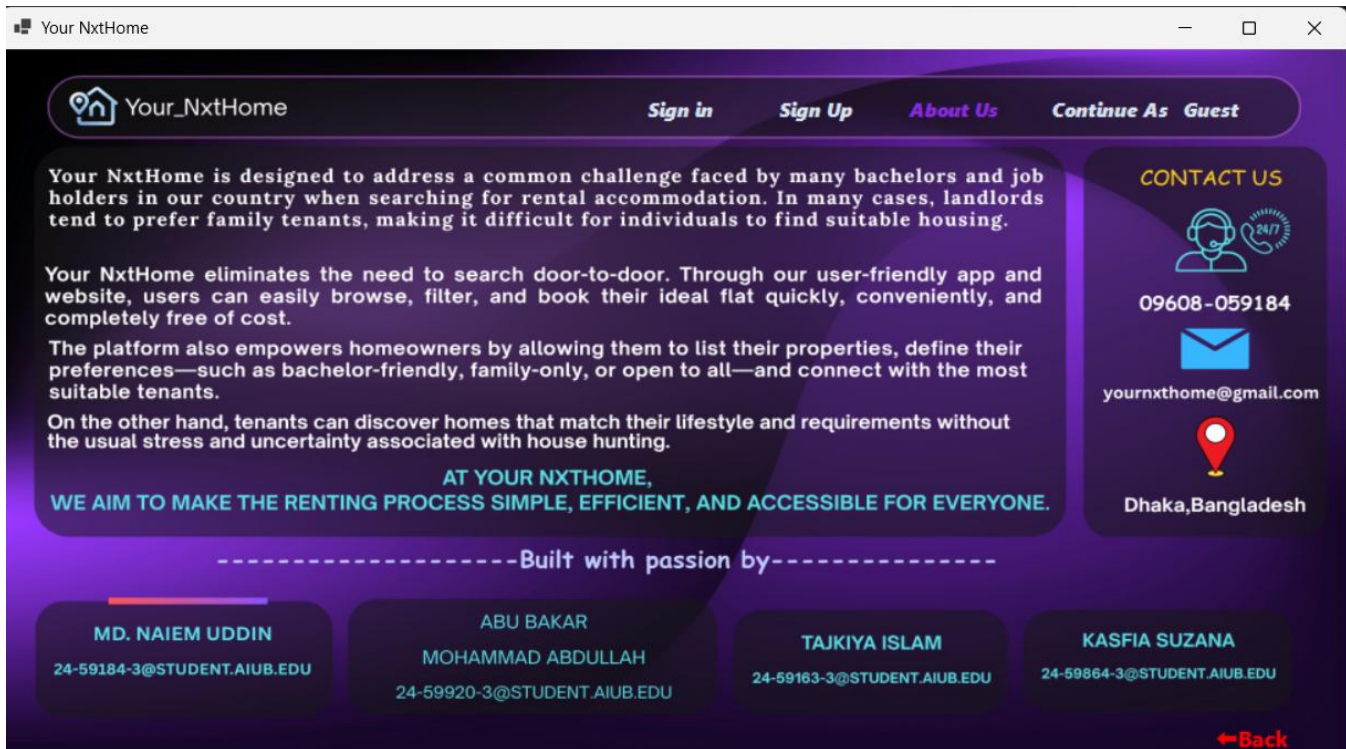
Add Update Delete Clear Back

ONLY
DEVELOPER CAN
CHANGE ROLE

GUEST MOOD (VIEW ONLY)



ABOUT US



Section 5. Conclusion

The Your NxtHome project successfully provides a smart and user-friendly rental management system that simplifies the process of finding and managing rental accommodations. The platform helps tenants and property owners communicate efficiently through features such as property listing, booking management, wishlist, messaging, and ratings.

The system reduces the difficulties of traditional house hunting by creating a centralized platform where users can search and filter properties according to their needs. Role-based access for Users, Admins, and Developers ensures proper system management and security.

The project also demonstrates effective use of C# .NET, Windows Forms, and Microsoft SQL Server for building a real-world database-driven application. Overall, the system improves transparency, reduces user effort, and provides a faster and more organized rental experience.

Section 6. Future Improvements:

Although **Your NxtHome** provides a complete rental management solution, several advanced features can be implemented in the future to improve usability, security, and user experience.

◆ Planned Future Enhancements

- **Property Availability Status**

Currently, multiple users can request booking for the same property. In future updates, once a booking is confirmed, the property status will automatically change to **Booked/Unavailable**, preventing other users from booking the same property.

- **Multiple Image Upload**

At present, users can upload only one image per post. Future versions will support **multiple property photos** for better visualization and detailed property presentation.

- **Live Map Location Sharing**

Property owners will be able to attach **Google Maps/live location** with their posts, making it easier for tenants to locate properties accurately.

- **Pet Policy Information**

A new option will be added to specify whether **pets are allowed or not** in the property.

- **Parking Availability**

Future posts will include parking-related information such as:

- Car parking available or not
- Bike parking facilities

- **User Verification System**

Additional verification methods will be implemented after registration to ensure trusted and verified users on the platform.

- **Call & Improved Communication Features**

The messaging system may be upgraded with:

- Audio/Video calling features
- Notification system

- **Scam Alert & Safety System**

A reporting and scam detection system may be added to identify suspicious users and improve platform safety.

- **Mobile Application Support**

Future development may include dedicated Android/iOS applications for better accessibility.

- **Online Payment Integration**

Digital payment gateways may be integrated for booking payments and advance rent transactions.

- ◆ **Final Vision**

The long-term goal of **Your NxtHome** is to become a complete smart rental ecosystem where tenants and property owners can communicate, verify, negotiate, and complete the entire renting process securely and efficiently from a single platform.